

Incomplete reversibility of fixation-induced changes in postmortem human brain white matter MRI following rehydration



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Background and Objective

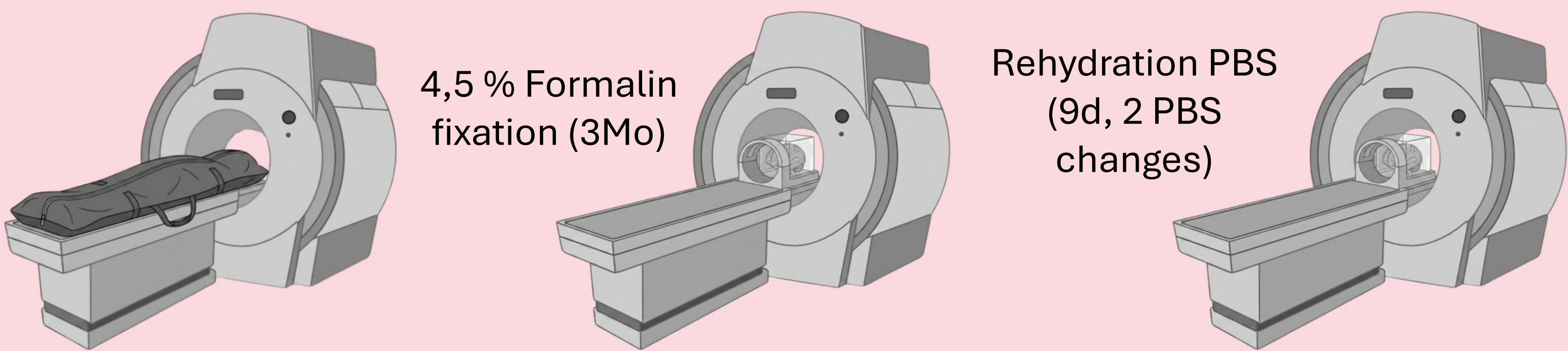
- Validation of MRI biomarkers: based on formalin-fixed ex situ imaging and histology
- Fixation: alters relaxation properties and tissue anisotropy¹⁻⁴
- Rehydration: suspected to partially reverse fixation effects
→ irreversible protein cross-linking suggests incomplete reversibility^{4,5}

Objective: To assess effects of formalin fixation and rehydration on MRI derived measures in human brain white matter

Methods

Subjects & MRI:

- Two human brains scanned on Siemens Prisma 3T



MRI Protocol:

- MP2RAGE: structural imaging and tissue segmentation
- Multi-echo GRE (12 echoes): R_2^* mapping
- Multi-shell DTI ($b = 0, 1000, 2000, 3000 \text{ s/mm}^2$)

Image processing:

- HD-BET⁶ - brain extraction, FSL FAST⁷⁻¹⁰ - segmentation, FSL eddy_correct⁷⁻⁹, FSL DTIFIT⁷⁻⁹ - mean diffusivity (MD), fractional anisotropy (FA), axial diffusivity (AD), radial diffusivity (RD)

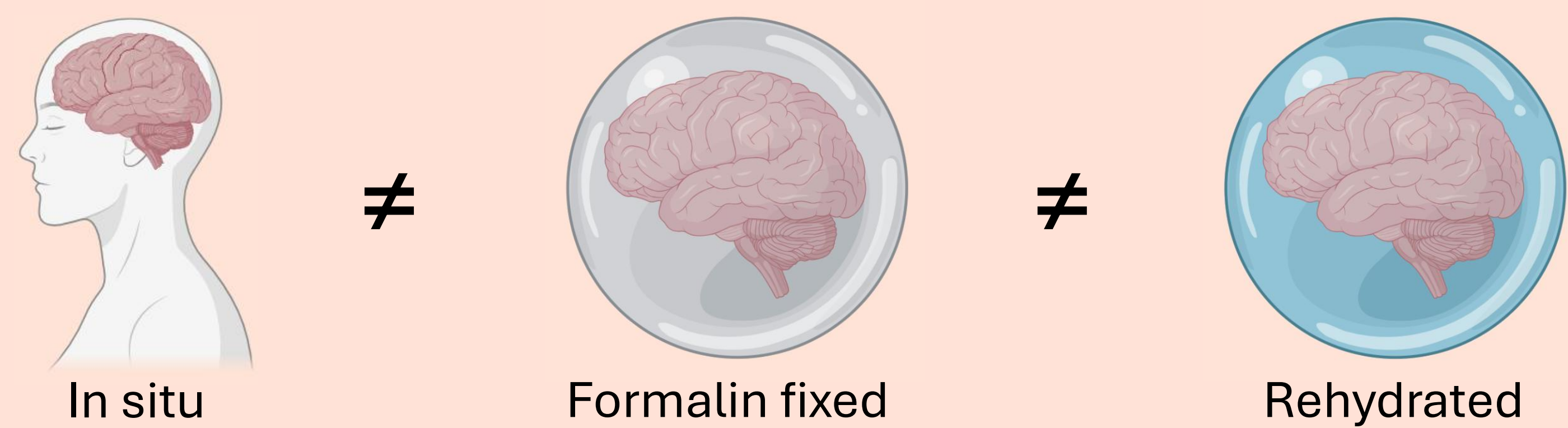
Analysis:

- Fiber angles derived from DTI V1, binned $18 \times 5^\circ$ ($0^\circ - 90^\circ$)
- R_2^* calculated voxel-wise, data normalised and registered
- R_2^* anisotropy index $AI = (R_2^{*max} - R_2^{*min}) / (R_2^{*max} + R_2^{*min})$

Conclusion

Fixation and rehydration:

- Substantial alterations in MRI-derived measures
- Heterogeneous & partial reversibility



Comparison of in vivo MRI to ex situ measurements for biomarker validation is limited

Results and Discussion

- Formalin fixation alters WM microstructure
- Fixation: FA $\downarrow$, MD $\downarrow$, AD $\downarrow$, RD $\downarrow$, AI $\downarrow$, R_2^* $\uparrow$
- Rehydration: FA $\downarrow$, MD $\uparrow$, AD $\uparrow$, RD $\uparrow$, AI $\uparrow$, R_2^* $\downarrow$
- Angular dependency: in situ $>$ rehydrated $>$ fixed

Table 1: Diffusion parameters FA, MD, AD and RD, as well as R_2^* and anisotropy index across the three tissue conditions.

	In situ	PM ex situ formalin-fixed	PM ex situ rehydrated
FA	0.5 ± 0.0	0.4 ± 0.1	0.3 ± 0.0
MD [$10^{-5} \text{ mm}^2/\text{s}$]	16.2 ± 6.5	13.0 ± 1.3	15.3 ± 2.8
AD [$10^{-5} \text{ mm}^2/\text{s}$]	24.8 ± 9.8	17.5 ± 2.2	19.9 ± 4.0
RD [$10^{-5} \text{ mm}^2/\text{s}$]	12.1 ± 4.7	11.0 ± 1.1	13.0 ± 2.3
R_2^* [s^{-1}]	27.8 ± 4.5	32.4 ± 0.4	28.4 ± 0.4
AI [%]	9.3 ± 3.3	1.5 ± 0.0	2.8 ± 0.4

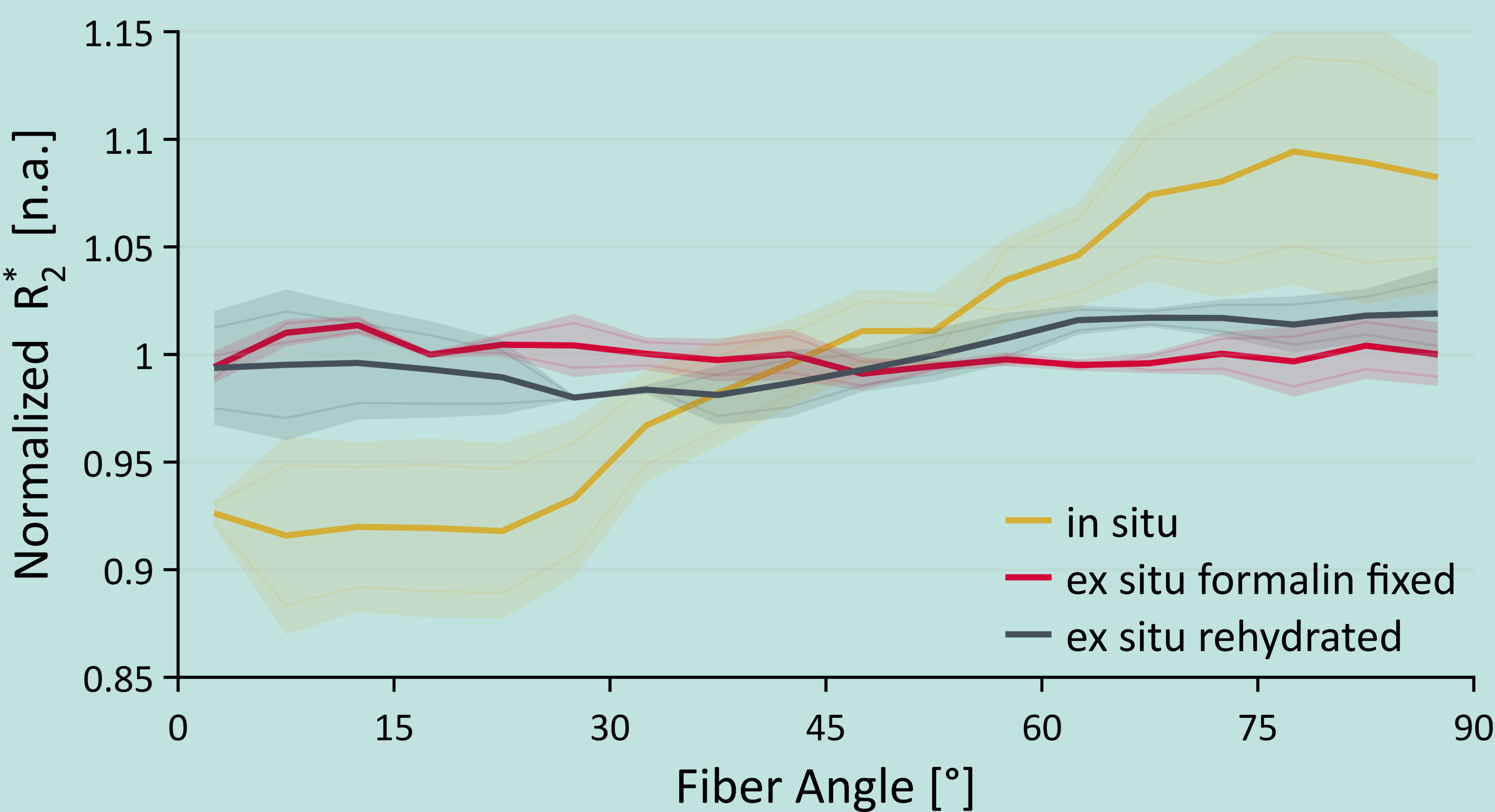


Figure 1: Normalized R_2^* values across the three tissue conditions over the calculated tissue fiber angle.

- Formalin fixation:**
 - FA, MD, AD, RD and R_2^* orientation dependency: $\downarrow$
- Rehydration:**
 - Partial recovery of MD, AD, AI and R_2^*
 - FA remained reduced, RD exceeded in situ
 - Tissue properties not fully restored

Key Message: Fixation effects are partly reversible, but not fully recoverable through rehydration

- Osmotic shifts & water distribution may cause apparent («pseudo-») recovery
- Fixation-induced changes should be considered when validating in vivo biomarkers using ex situ MRI and histology

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